PAX SERVICES GROUP

PAXSEAL™ BUILDING LIFECYCLE PLAN

Roof Lifecycle Cost Analysis

MARKET PROJECTION
Typical school

Culbertson (N. Virginia) market
October 14, 2026

Planned maintenance is projected to save the owner of a typical Culbertson (N. Virginia) school $1.62M over the next 20 years.

About 37% less than reactive repair: $1.20M lower spending plus $418K more roof life at year 20. Range $851K–$2.11M from conservative to upside assumptions; $843K in today's dollars at a 7% discount rate.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE · CONDITION	NEW ROOF TODAY
Culbertson (N. Virgi…	K-12	120,000 sq ft	Modified bitumen	8 yrs · Good	$1.14M

PAXSEAL NET COST
$2.76M
$2.65M spent + $684K roof used − $570K roof left

REACTIVE NET COST
$4.38M
$3.85M spent + $684K roof used − $152K roof left

REPAIR & STORM SAVINGS
$1.88M
Unbudgeted repairs and storm damage avoided

ROOF PRICES
2× in 15 yrs
At 4.6%/yr, the long-run trend in commercial roofing costs

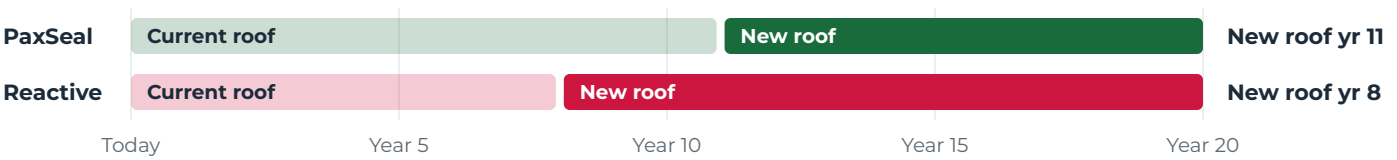
Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal fees	$441K	—	**−$441K**
Repairs paid by owner	$278K	$2.13M	**+$1.85M**
Expected storm damage	$57K	$86K	**+$29K**
Roof replacements	$1.87M	$1.63M	**−$240K**
Cash spending	**$2.65M**	**$3.85M**	**+$1.20M**
Current roof value used	$684K	$684K	—
Roof life left at year 20	−$570K	−$152K	**+$418K**
Net 20-year cost	**$2.76M**	**$4.38M**	**+$1.62M**

What this means

1. **A later replacement.** The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
2. **Smaller repairs.** Emergency repairs run about 3.9× the planned rate in a school, and the PaxSeal fee covers half of planned repairs. Together that saves $1.85M.
3. **A predictable budget line.** Emergency repairs become a fixed annual cost the board can plan for, and leaks stop closing classrooms.
4. **The program pays for itself.** Each $1 of fees avoids about $3.73 of other cash roof costs.

Local risk profile

Prince William County, VA (51153)

Thunderstorm wind / derecho, snow and ice

Strong wind and winter weather are both rated Relatively High by FEMA, and tropical storm remnants arrive about once every 12 years. Expected storm damage to this roof over 20 years is **$57K** if maintained and **$86K** with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

Hail	**Relatively Moderate**
Strong wind	**Relatively High**
Tornado	**Relatively Moderate**
Hurricane	**Relatively Moderate**
Winter weather	**Relatively High**
Ice storm	**Relatively Low**

FEMA National Risk Index rating

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	**20 / 15 yrs**	Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties)
Roof cost growth	**4.6%/yr**	BLS Producer Price Index, nonresidential roofing, 2007–2025
Replacement cost	**$9.50/sq ft**	Culbertson (N. Virginia) contractor price ranges
Reactive repair cost	**3× planned × 1.3**	Industry estimates; ×1.3 for leak damage (school). PaxSeal fee covers half of planned repairs
Storm damage	**0.15%/yr**	Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index
PaxSeal fee	**$16,400/yr**	Year 1, rising 3%/yr. Estimate; your proposal states the actual fee

Next step: a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email info@paxserv.com.

Estimates, not a quote. Projections from Culbertson (N. Virginia) market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance.